

OIR Hub: Experimental Design for a 專題 Research Contribution

Turning a deployed bilingual RAG system into three falsifiable experiments

Author: David · Department of Computer Science, Tunghai University (東海大學)
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Status: Working system deployed to LINE; document-routing LLM implemented; no measured baseline yet
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0. Summary — the one page to read before the meeting

The system built so far is a *deliverable*, not a *study*. A 專題 needs three things a website does not have: a question that could come out either way, a measurement that decides it, and a comparison against something simpler. The existing code already contains the material for all three — what is missing is an annotated evaluation set, which is the single blocking dependency for every number in this document.

Three experiments are proposed, in descending order of research value per unit of effort:

#	Experiment	Core question	Primary metric	Role
1	Semantic supersession routing UPDATE / NEW / REDUNDANT	Can an LLM judge decide that a new bulletin <i>replaces</i> an existing one, where similarity and timestamps cannot?	Macro-F1 over 3 classes; target-selection accuracy for UPDATE	CORE the paper's contribution
2	Cross-lingual retrieval ablation EN ↔ zh-TW	How much retrieval quality is lost when query language ≠ document language, and what recovers it?	Recall@k, nDCG@10, per-stage latency	SUPPORTING system justification
3	Proficiency-adapted answering CEFR / TOCFL control	Does controlling answer readability improve comprehension without losing factual grounding?	Readability conformity, citation-grounding rate, human comprehension	STRETCH offer, do not promise

THE ONE SENTENCE TO OPEN THE MEETING WITH

“The website is the platform, not the 專題. The 專題 is this: current RAG systems assume the document collection is static, but the OIR's is not — bulletins are superseded every semester. I want to measure whether an LLM can detect supersession where embedding similarity and timestamps fail, and I need roughly three weeks with 宜芳姐 to build the labeled set before any of it produces numbers.”

1. Why the current deliverable was rejected, stated precisely

The rejection is not a judgement on the engineering. It is a category problem. The table below is the difference, and it is worth showing to the advisor directly, because it demonstrates that the gap is understood rather than merely conceded.

What exists now (a system)	What a 專題 requires (a study)
Two websites that work	A hypothesis that could turn out false
A chatbot that answers questions	A dataset with ground-truth labels
A routing LLM that makes decisions	Baselines the method must beat
“It seems to work well”	Macro-F1 = 0.81 vs 0.63, $p < 0.01$, $n = 150$
Screenshots	An ablation table and an error analysis

Every row on the right is reachable from what is already built. No new system needs to be written — the work shifts from building to measuring.

2. The research gap, and why it is real

Retrieval-augmented generation was introduced as a way to attach a non-parametric knowledge store to a language model (Lewis et al., 2020). Almost all subsequent work optimises retrieval *quality* over a corpus that is treated as fixed. Corpus *maintenance* — deciding what enters the index, what it replaces, and what should be discarded — is generally handled as engineering, not research.

Where maintenance is addressed in practice, the mechanisms are deterministic: last-modified timestamps, content hashes, or version-control history. These detect that a file *changed*. They cannot detect that document B *supersedes* document A when A and B are different files from different sources with no shared identifier — which is exactly the OIR case, where the 2027 exchange-program bulletin arrives as a fresh PDF attachment rather than an edit to the 2026 one.

Recent work has begun to measure the consequences. Benchmarks of knowledge conflict in RAG now treat freshness conflicts — outdated evidence retrieved alongside current evidence — as a distinct and poorly handled category, and find that models frequently fail to resolve them appropriately. Work evaluating rerankers under evolving knowledge reports a consistent failure mode in which rerankers favour semantically rich but obsolete passages over temporally valid ones, with obsolete-preference rates remaining high across nearly the entire model spectrum. The diagnosis offered is that standard retrieval training rewards semantic overlap and gives almost no supervision for temporal validity.

There is also an active disagreement worth entering. One line of argument holds that freshness should be resolved deterministically by metadata rules and that asking an LLM to judge it is an architectural mistake; the counter-position is that in domains where supersession is semantic rather than timestamped, deterministic rules have nothing to key on. The OIR corpus is a clean instance of the second case, and a small honest experiment is a legitimate contribution to that argument.

POSITIONING STATEMENT

Prior work measures the *downstream damage* caused by stale documents at query time. This project addresses the problem one stage earlier — at ingestion — by treating supersession detection as a supervised classification task over a real institutional corpus, and asking whether semantic judgement beats the deterministic heuristics currently deployed in production RAG pipelines.

Honest scope limit to state aloud: this is a single-domain study on one institution's corpus in two languages. It is a case study with baselines, not a general benchmark. That is appropriate for an undergraduate 專題 and should be framed as such rather than overclaimed.

3. Experiment 1 — Semantic supersession routing CORE

RESEARCH QUESTION

Given an incoming OIR document and an existing indexed corpus, can an LLM judge classify the document's relationship to the corpus (**UPDATE** / **NEW** / **REDUNDANT**) more accurately than similarity-threshold and metadata heuristics — and if so, on which cases does the advantage appear?

H1. Similarity-based methods fail systematically on *near-duplicate-but-distinct* pairs (two different scholarships sharing a template) and on *distant-but-superseding* pairs (a reworded successor bulletin). **H2.** An LLM judge supplied with retrieved candidate documents outperforms both threshold baselines and a zero-shot LLM judge on macro-F1. **H3.** The gain is concentrated in the UPDATE class; NEW is easy for all methods.

3.1 Class definitions and labeling rubric

This rubric is the most important artifact in the entire project. It must be agreed with 宜芳姐 *before* labeling starts, written down, and included in the paper's appendix. Ambiguous class boundaries are the most common reason undergraduate datasets produce meaningless numbers.

Label	Definition	OIR example
UPDATE	Document describes the same program/policy as an existing document, and the existing one is now invalid or partially invalid. The old document must be retired or marked superseded.	2027 Spring exchange application opens, replacing the 2026 Spring bulletin: same partner universities, new deadlines, revised GPA requirement.
NEW	Document describes a program, policy or announcement not currently covered. The existing corpus remains fully valid.	A newly signed MOU with a Japanese partner university; a first-time summer research scholarship.
REDUNDANT	Document adds no retrievable information beyond what is indexed — a repost, a reformatted copy, or a trivial edit.	The same scholarship notice reposted with a corrected office phone number or a new header image.

Edge cases to decide in advance (write the ruling into the rubric)

- **Partial supersession.** A bulletin that changes only the deadline of an existing program. Proposed ruling: UPDATE, with a secondary field `scope = partial`.
- **Split.** One old bulletin superseded by two new ones. Proposed ruling: UPDATE against the old document, allowing multiple incoming documents to target the same parent.
- **Annual recurrence with no change.** Same content, new academic year stamp. Proposed ruling: UPDATE, not REDUNDANT — the validity window moved even though the text did not.
- **Language pair.** The zh-TW original and its English translation of the same bulletin. Proposed ruling: exclude from the routing set entirely; handle as a linking problem, and note the exclusion in the paper.

3.2 Dataset construction protocol

1. **Collect** the OIR bulletin archive from 宜芳姐 — target at least three academic years so that genuine supersession chains exist. Record for each document: source, date received, language, and program identifier if one exists.
2. **Sample** 150–200 incoming-document cases, stratified to approximately 40% UPDATE, 40% NEW, 20% REDUNDANT. Natural distribution will be heavily skewed toward NEW; oversample the minority classes deliberately and report both the natural and sampled distributions.
3. **Include hard negatives on purpose.** At least 25 cases must be template-sharing but genuinely distinct programs (high lexical overlap, correct label NEW). These are the cases that discriminate between methods; without them every method scores well and the experiment proves nothing.
4. **Double-annotate** a 60-item subset with a second annotator (a teammate, or 宜芳姐 plus yourself). Report **Cohen's κ** . If $\kappa < 0.6$, the rubric is broken — revise it and re-annotate rather than proceeding.
5. **Split** into dev (50) for threshold tuning and prompt iteration, and test (100+) touched only once, at the end. State clearly in the paper that thresholds were tuned on dev only.
6. **Store** as versioned JSONL with a fixed schema: `doc_id`, `text`, `lang`, `date`, `gold_label`, `gold_target_id` (null unless UPDATE), `annotator`, `notes`.

3.3 Conditions to compare

ID	Method	Purpose in the argument
B0	Majority-class predictor	Floor. Shows what the class imbalance alone buys.
B1	Metadata rule: recency + program-ID string match	Represents the deterministic production approach. Expected to be strong where IDs exist, blind where they do not.
B2	Cosine similarity over e5 embeddings, threshold tuned on dev	The naive semantic baseline. H1 predicts it cannot separate the two hard case types with any single threshold.
B3	BM25 lexical overlap, threshold tuned on dev	Lexical control. Templates inflate this; a useful contrast against B2.
B4	bge-reranker-v2-m3 cross-encoder score, threshold tuned on dev	Stronger pairwise semantic scorer without generation. Isolates how much of any LLM gain is really just better pair scoring.
M1	LLM judge, zero-shot, no retrieval context	Tests whether the LLM can judge at all without seeing the corpus.
M2	LLM judge + top-5 retrieved candidates, structured JSON output	The proposed method. Current implemented system.
M3	M2 + few-shot exemplars drawn from dev	Cheap ablation; isolates the contribution of in-context examples.
M4	Hybrid cascade: B1 decides when confident, M2 otherwise	The practical proposal. Often the best accuracy-per-token result, and an honest engineering conclusion.

Run each LLM condition with `temperature = 0` and at least 3 repeats; report mean and standard deviation. Non-determinism across repeats is itself a reportable finding, especially for a 4B model.

3.4 Metrics

- **Macro-F1** across the three classes — the headline number. Macro, not micro, because REDUNDANT is the smallest and most interesting class.
- **Per-class precision and recall**, plus the full 3×3 confusion matrix for every condition.
- **Target-selection accuracy** — restricted to gold-UPDATE cases, did the method identify the correct document to supersede? A method can label UPDATE correctly and still point at the wrong parent; this is a separate and more demanding metric.
- **Cost**: mean latency per decision (wall clock) and mean tokens generated per decision.
- **Significance**: McNemar's test for the paired M2-vs-best-baseline comparison; bootstrap 95% confidence intervals (2,000 resamples) on macro-F1. With $n \approx 150$ this is essential — do not report a bare point estimate.

3.5 Error analysis — the section that makes it a paper

Numbers alone are a lab report. The contribution lives in explaining *why* the baselines fail. Plan to produce:

- A scatter plot of cosine similarity against gold label, showing directly that no horizontal threshold separates UPDATE from NEW. If H1 is right, this single figure is the strongest thing in the paper.
- A qualitative taxonomy of LLM failure modes — hallucinated supersession, missed partial updates, over-eager REDUNDANT — with two or three real anonymised examples each.
- A language breakdown: does routing accuracy differ for zh-TW versus English documents? This links Experiment 1 to Experiment 2 and unifies the paper.

IF THE HYPOTHESIS FAILS

If B1 or B2 matches the LLM judge, that is a publishable result, not a disaster — it would be independent confirmation of the deterministic-recipe position, on a new domain and a new language pair. Say this to the advisor explicitly. A student who has pre-committed to reporting a negative result is demonstrating research maturity, and it removes the incentive to quietly tune the experiment toward the desired answer.

4. Experiment 2 — Cross-lingual retrieval ablation SUPPORTING

RESEARCH QUESTION

For an EN/zh-TW institutional corpus, how much retrieval quality is lost when the query language differs from the document language, is the loss symmetric in both directions, and which pipeline components recover it at what latency cost?

Existing cross-lingual RAG evaluations report that mismatched query and document languages consistently underperform same-language retrieval, and — importantly — that the degradation is *directional*: in an Arabic–English study, English queries against Arabic documents degraded sharply while the reverse direction did not degrade significantly for one encoder, and the multilingual-e5 family degraded substantially in both directions. Work on multilingual retrieval benchmarks has similarly found that encoders scoring well in English can collapse on another language, a failure invisible to English-only evaluation. Whether the same asymmetry holds for English ↔ Traditional Chinese on short institutional notices is an open and locally relevant question.

4.1 Evaluation set

- **80–120 questions** with gold relevant passages annotated. Sources: real anonymised LINE query logs (highest value — they are genuinely representative), plus the OIR's existing FAQ, plus questions written by 宜芳姐 covering topics students ask in person.
- **Stratify by direction**: EN query → zh-TW document; zh-TW query → EN document; EN → EN; zh → zh. Roughly equal cells.
- **Stratify by difficulty**: single-passage answerable, multi-passage answerable, and unanswerable (no supporting document exists). The unanswerable slice supports an abstention analysis and is often the most revealing.
- Annotate *all* relevant passages per query, not just one, or Recall@k is undefined and nDCG is wrong.

4.2 Ablation grid

Factor	Levels	What it isolates
e5 asymmetric prefixes	with / without	Sanity check — a known result. Report it as pipeline validation, not as a finding.
Encoder	multilingual-e5-base / BGE-M3	Whether a stronger multilingual encoder closes the cross-lingual gap on this corpus.
Retrieval mode	dense / BM25 / hybrid with RRF fusion	Whether lexical signal helps or hurts across a script boundary. Chinese needs proper segmentation for BM25 — use CKIP or jieba and say which.
Reranking	off / bge-reranker-v2-m3 top-30	The main recovery mechanism, and the main latency cost.
Query rewriting	off / LLM rewrite / translate-to-corpus-language	Tests whether translation still beats native cross-lingual matching, as several studies suggest.

Full factorial is $2 \times 2 \times 3 \times 2 \times 3 = 72$ runs. Do not run all of them. Fix a sensible default configuration and vary one factor at a time (one-factor-at-a-time ablation, ~12 runs), then run a small full factorial over only the two factors that mattered most.

4.3 Metrics

- **Recall@1 / @5 / @10, nDCG@10, MRR** — reported separately per language direction, never averaged across directions, since the whole point is the asymmetry.
- **Cross-lingual gap**: the drop from same-language to cross-language performance, reported as a single percentage per encoder. This is the number the section is about.
- **Per-stage latency** using `time.perf_counter()` instrumentation around embed / search / rerank / generate, so the accuracy-versus-latency tradeoff table is real rather than estimated.
- **Answer-level effect**: end-to-end answer correctness on a subset, to show that retrieval gaps actually propagate to what the student reads.

5. Experiment 3 — Proficiency-adapted answering STRETCH

RESEARCH QUESTION

Can the answer generator reliably hit a target proficiency level (CEFR for English, TOCFL/HSK band for Chinese), and does doing so improve comprehension for international students without reducing factual grounding?

Recent evaluations of CEFR-controlled generation find that structured prompting with explicit lexical constraints produces markedly better proficiency alignment than unconstrained prompting, which controls readability only weakly. Instruction-tuned approaches to readability control likewise report that general-purpose models follow readability instructions only partially, with error growing at the extremes. Almost all of this work is English-centric or single-language; a bilingual EN/zh-TW setting with a real user population is largely unexplored, and Traditional Chinese has a Taiwan-developed readability toolchain (CRIE and its CFL variant) that makes the Chinese side measurable rather than hand-waved.

5.1 Conditions

1. **C0 — Default.** Current RAG answer, no readability instruction.
2. **C1 — Level-prompted.** Target level stated in the prompt ("answer at CEFR B1").
3. **C2 — Level + lexical constraint.** Target level plus an explicit allowed-vocabulary list, the condition prior work finds strongest.
4. **C3 — Two-stage.** Generate normally, then simplify in a second pass. More faithful, slower, more tokens — the tradeoff is the point.

5.2 Automated evaluation (do this part regardless)

- **English:** readability indices (Flesch–Kincaid, or a CEFR classifier), type-token ratio, mean sentence length, proportion of tokens outside the target CEFR wordlist.
- **Chinese:** CRIE / CRIE-CFL feature extraction, proportion of words above the target TOCFL band, mean sentence length in characters. Use a Taiwan-trained segmenter (CKIP) rather than a Mainland-default one.
- **Control error:** absolute distance between target level and measured level, averaged. This is the headline metric and it requires no human subjects.
- **Grounding:** citation-support rate — the proportion of factual claims in the answer traceable to a retrieved passage. Critical: simplification must not be allowed to quietly drop the deadline or the eligibility condition. Track `chat_citations` in the relational schema for this.

5.3 Human evaluation (only if time allows)

- 20–30 international students recruited through the OIR, within-subject design, answer order counterbalanced.
- Per item: a 1–5 comprehension rating, a 1–5 usefulness rating, and one factual comprehension check question with a verifiable answer. The check question matters — self-reported comprehension is unreliable on its own.
- Collect self-reported proficiency (CEFR band, TOCFL level) to test whether adaptation helps lower-proficiency students more, which is the interesting interaction.
- **Administrative dependency:** confirm with the advisor whether the department requires an ethics/consent process for human subjects, and budget two weeks for it. This is the most common reason a student user study never happens.

6. Timeline to December 2026

Period	Work	Concrete output
Weeks 1–3 Sept–early Oct	Write the labeling rubric. Meet 宜芳姐. Collect the bulletin archive. Label 150–200 routing cases. Double-annotate 60, compute κ .	Frozen rubric; <code>routing_gold.jsonl</code> ; reported κ
Weeks 2–4 parallel	Build the 80–120 item retrieval gold set from LINE logs and the FAQ, stratified by language direction.	<code>retrieval_gold.jsonl</code>
Weeks 4–6	Build the evaluation harness: one script, one config file, one results CSV. Implement B0–B4 and M1–M4. Tune thresholds on dev only.	Reproducible <code>eval.py</code> ; dev results table
Weeks 6–8	Run Experiment 1 on the held-out test set. McNemar tests, bootstrap CIs, confusion matrices, the similarity-vs-label scatter figure.	Experiment 1 results; 3 figures
Weeks 8–10	Run Experiment 2 ablation grid with per-stage timing. Produce the accuracy-versus-latency table.	Experiment 2 ablation table
Weeks 10–12	Error analysis and failure taxonomy. Experiment 3 automated metrics if on schedule.	Error taxonomy; optional Exp. 3 table
Weeks 12–14	Write the paper. Prepare the defense. Human study only if everything above is complete.	Draft paper; 專題 report; slides

7. Risk register

Risk	Severity	Mitigation
Not enough real supersession pairs exist in the archive	Critical	Verify the archive spans ≥ 3 years <i>in week 1</i> , before committing to the framing. Fallback: augment with a second institution's public bulletins, or construct a controlled synthetic subset and report it separately from the natural subset.
宜芳姐 lacks time to annotate	High	Ask for 3 hours total, not open-ended help. Pre-label everything yourself first; her role becomes review and adjudication of disagreements, which is far cheaper.
Low inter-annotator agreement	High	The rubric edge cases in § 3.1 exist to prevent this. If $\kappa < 0.6$, revise and re-annotate — do not proceed on a broken rubric.
4B local model too weak to show any effect	Medium	Model size becomes a reported variable rather than a confound: run one condition against a larger model via API and report the gap. "What can an 8GB laptop GPU actually do here" is a legitimate and useful framing.
Scope creep into building more features	Medium	Feature freeze on the system from week 1. Nothing new is built unless an experiment requires it. This is the most likely way the project fails.
Ethics approval blocks the human study	Low	Experiment 3's automated half needs no human subjects. Keep the human study strictly optional.

8. Meeting agenda for this evening

Structure the conversation in this order. Leading with the concession buys credibility for everything after it.

1. **Concede the point first, precisely.** "You are right that a website is not a 專題. What it lacks is a hypothesis, a labeled dataset, and baselines." Do not defend the websites.
2. **Reframe what already exists.** The routing LLM is not a feature — it is an unevaluated method for an under-studied problem. Show the gap in § 2.
3. **Present the three experiments** using the table in § 0, and say plainly which one is the core.
4. **Name the bottleneck before the advisor does.** Three weeks of annotation with 宜芳姐 come before any coding. Ask for help securing her time — an advisor's email carries weight a student's does not.
5. **Ask the advisor to choose.** Which experiment is the core, and is the scope right for one semester? This converts a judge into a co-owner of the plan.

Questions to ask, and why each one matters

- "Is a single-institution case study acceptable scope for 專題, or does the department expect a general benchmark?" — settles the overclaiming risk immediately.
- "If the LLM does *not* beat the baselines, is a negative result acceptable for the report?" — the answer determines how much risk can be taken, and asking it signals research maturity.
- "Does the department require an ethics process for a student survey?" — decides whether Experiment 3 is live or dead.
- "Would you be willing to co-author if this reaches workshop quality?" — an advisor with authorship stake gives far better

supervision, and a co-authored workshop paper is the single strongest artifact for a graduate application.

- "Which of the three would you want to see first?" — closes the meeting with the advisor holding the decision.

What to bring

This document, printed. The class-definition table from § 3.1. One sample OIR bulletin pair that is a genuine UPDATE, and one pair that looks nearly identical but is genuinely NEW. Those two concrete examples do more persuasive work than any slide — they make the research problem visible in ten seconds.

9. References

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